

JINZHUANG DOU, Ph.D.

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RESEARCH INTERESTS

Statistical Genomics, Computational Biology, AI in Biomedicine

EDUCATION

Ph.D in Genetics

08/2012 – 06/2015

Ocean University of China, QingDao, China

M.S. in Applied Mathematics

08/2009 – 06/2012

Ocean University of China, QingDao, China

B.S. in Applied Mathematics

08/2005 – 06/2009

Ocean University of China, QingDao, China

RESEARCH EXPERIENCE

Assistant Professor

06/2025 -

The University of Alabama at Birmingham, Birmingham, AL, USA

Data Scientist

01/2021 -06/2025

MD Anderson Cancer Center, Houston, TX, USA

Postdoctoral research fellow

07/2018 – 01/2021

MD Anderson Cancer Center, Houston, TX, USA

Postdoctoral research fellow

07/2015 – 07/2018

Genome Institute of Singapore, A*STAR, Singapore

AWARDS

“Wall of Science”, MD Anderson Cancer Center (2023)

MAJOR PUBLICATIONS (* co-first author; ^ corresponding author)

1. Z Wang, K Chen, **J Dou**[^] (2026). Single-Cell Indel Detection Enhances Genetic Ancestry and Cellular Lineage Analysis. *Bioinformatics* (submitted).
2. S, R Luo, Z Wang, J Dou, K Chen, R Chen (2026). Somatic variant detection in normal tissues from single cell sequencing data. *Cell Systems* (Accepted in Principle)
3. **J Dou**, Y Tan, KH Kock, J Wang, X Cheng, LM Tan, KY Han, CC Hon, WY Park, JW Shin, H Jin, H Chen, L Ding, S Prabhakar, N Navin. K Chen (2023). Monopogen: single nucleotide variant calling from single cell sequencing. *Nature Biotechnology*. 2023 Aug 17:1-0 [Highlight in *Nature Genetics*. <https://www.nature.com/articles/s41588-023-01544-2>]
4. D Wu*, **J Dou***, X Chai*, C Bellis, A Wilm, C.C. Shih, W.W.J Soon, N Bertin, C.B Lin, C.C Khor, M DeGiorgio, S.S Cheng, L Bao, N Karnani, W Hwang, S Davila, P Tan, A Shabbir, A Moh, E Tan, J.N Foo, L.L Goh, K.P Leong, R.S.Y Foo, C Lam, A Richards, C.Y Cheng, T Aung, T Wong, H Ng, M Ackers Johnson, E Aliwarga, K Kim Ban, D Bertrand, J Chambers, D Hui Chan, C Li Chan, M Chee, M Chee, P Chen, Y Chen, E Chew, W Chew, L Chiam, J Chong, I Chua, S Cook, W Dai, R Dora Joo, C Foo, R Goh, A Hillmer, I Irwan, F Jaufeerally, A Javed, J Jeyakani, J Koh, J Koh, P Krishnaswamy, J Kuan, N Kumari, A Lee, S Lee, S Lee, Y Lee, S Leong, Z Li, P Li, J Liew, O Liew, S Lim, W Lim, C Lim, T Lim, C Lim, S Loh, A Lok, C Chin, S Majithia, S Maurer-Stroh, W Meah, S Mok, N Nagarajan, P Ng, S Ng, Z Ng, J Ng, E Ng, S Ng, S Nusinovici, C Ong, B Pan, V Peder Gnana, S Poh, S Prabhakar, K Prakash, I Quek, C Sabanayagam, W See, Y Sia, X Sim, W Sim, J So, D Soon, E Tai, N Tan, L Tan, H Tan, W Tan, M Tandiono, A Tay, S Thakur, Y Tham, Z Tiang, G Toh, P Tsai, L Veeravalli, C Verma, L Wang, M Wang, W Wong, Z Xie, Yeo, L Zhang, W Zhai, Y Zhao, J Liu, C Wang (2019) Large-scale whole-genome sequencing of three diverse Asian populations in Singapore. *Cell* 179(3):736-749 [Cover story]

5. Li*, V Mohanty*, **J Dou***, Y Huang*, P P. Banerjee, Q Miao, J Lohr, T Vijaykumar, B Knoechel, R Basar, S Liang, M Kaplan, M Daher, E Liu, L Muniz-Feliciano, T J. Laskowski, D Marin, S Mielke, R E. C, Elizabeth J. Shpall, K Chen, K Rezvani (2023). Metabolic competition is an important driver of tumor resistance after CAR NK cell therapy and can be overcome by cytokine engineering. *Science Advance* 9(30), eadd6997.
6. **J Dou**, S Liang, V Mohanty, X Cheng, S Kim, J Choi, Y Li, K Rezvani, R Chen, K Chen (2022). Unbiased integration of single cell multi-omics data. *Genome Biology* 23(1), pp.1-25
7. **J Dou**, D Wu, L Ding, K Wang, M Jiang, X Chai, D. Reilly, E Tai, J Liu, X Sim, S Cheng, C Wang (2020). Using off-target data from whole-exome sequencing to improve genotyping accuracy, association analysis, and polygenic risk prediction. *Brief in Bioinformatics* bbaa084
8. **J Dou***, B Sun*, X Sim, I Irawan, J D Hughes, D F Reilly, E Tai, J Liu, C Wang (2017). Estimation of kinship coefficient in structured and admixed populations using sparse sequencing data. *PLoS Genetics* 13: e1007021s.
9. **J Dou***, H Dou*, C Mu, L Zhang, Y Li, J, T Li, X Hu, S Wang, Z Bao (2017). Whole-genome restriction mapping by “sub haploid”-based RAD sequencing: an efficient and flexible approach for physical mapping and genome scaffolding. *Genetics* 206:1237-1250. [Cover story]
10. **J Dou**, X Zhao, X Fu, W Jiao, N Wang, L Zhang, X Hu, S Wang, Z Bao (2012). Reference-free SNP calling improved accuracy by preventing incorrect calls from repetitive genomic regions. *Biology Direct* 7:17.
11. **J Dou**, X Li, Q Fu, W Jiao, Y Li, T Li, Y Wang, X Hu, S Wang, Z Bao (2016). Evaluation of the 2b-RAD method for genomic selection in scallop breeding. *Scientific Report* 6:19244.
12. W Jiao*, X Fu*, **J Dou***, H Li, H Su, J Mao, Q Yu, L Zhang, X Hu, X Huang, Y Wang, S Wang, Z Bao (2013). High-resolution linkage and quantitative trait locus mapping aided by genome survey sequencing: building up an integrative genomic framework for a bivalve mollusk. *DNA Research* 21 (1): 85-101. [Editor's choice]

ACTIVE GRANT AWARDS

UAB Comprehensive Diabetes Center Seed Funding	\$50,000	09/2026-09/2027
NextGen Scholar Grant (O'Neal Comprehensive Cancer Center; UAB)	\$50,000	06/2026-06/2028
Start-up Support for New Faculty Recruitment Package (UAB)	-	06/2025-

INVITED TALKS

1. ICIBM 2026 — 08/2026, Buffalo, NY; Genetic information inference from Single cell sequencing
2. UAB DBIDS Power Talk — 09/2025; Computational frameworks integrating single-cell data with GWAS to identify causal signals
3. ICIBM 2025 — 08/2025, Columbus, Ohio; Ancestry-informed single-cell genomics and computational modeling.
4. Texas Single Cell Seminar —10/2023, Zoom; Single nucleotide variant calling from single cell sequencing using Monopogen.
5. Texas Single Cell Seminar —05/2021, Zoom; Unbiased integration of single cell multi-omics data
6. CZI Seed Networks 2020 Annual Meeting—11/2020, Zoom; Unbiased integration of single cell multi-omics data
7. Texas Single Cell Seminar —02/2020, Houston, TX; Single-cell characterization of CAR-NK cell exhaustion in transplantation immunotherapy.

SOFTWARES

1. *CCAdge* (CCA-based DEGs analysis in single cell data). Identify biologically varied genes in population level single cell data.
<https://github.com/KChen-lab/CCAdge>
2. *Monopogen* (Single nucleotide variant calling from single cell sequencing). Achieve SNV calling from single cell RNA sequencing, single cell ATAC sequencing.
<https://github.com/KChen-lab/Monopogen>
3. *bindSC* (Bi-order integration of single cell multi-omics data). Achieve bi-order integration (in silico multi-omics data) of single cell RNA sequencing, single cell ATAC sequencing, spatial transcriptomics and CyTOF data.
<https://github.com/KChen-lab/bindSC>
4. *WEScall* (A genotype calling pipeline for whole exome sequencing (WES) data). Analyze both target and off-target data to facilitate WES studies with decent amount of off-target data.
<https://github.com/dwuab/WEScall>

5. *SEEKIN* (Sequencing-based estimation of kinship and inbreeding coefficients). Estimates the kinship and inbreeding coefficients based on extremely low-sequencing coverage datasets (typically lower than 0.5X).
<https://github.com/chaolongwang/SEEKIN>
6. *AMMO* (An integrated package for whole-genome restriction mapping and genome scaffolding). Generate 2b-RAD-based restriction maps for scaffolding draft genome assemblies produced by using short Illumina reads or long PacBio reads.
<https://github.com/jinzhuangdou/AMMO>
7. *RADtyping* (De novo RAD Genotyping in Mapping Populations). Accurate De Novo codominant and dominant RAD genotyping in mapping populations.
<https://github.com/jinzhuangdou/RADtyping>

TEACHING (^ quest lecture)

1. 01/2026-05/2026 INFO604/704: Next-Generation Sequencing Data Analysis
2. 08/2022 Multiomics integration and feature selection ^, Gulf Coast Consortia (GCC) Single Cell Omics Workshop
3. 11/2025 INFO601/701: Introduction to Bioinformatics^

TRAINING & MENTORSHIP

Graduate students

Ruofan Mao, MS in computer science from UAB, 08/2025-

Postdoctoral Fellows

Xin Liang, Ph.D. in bioinformatics from Harbin Medical University, 09/2025-

PEER REVIEW EXPERIENCE

Ad Hoc journal reviewer

Cell, Nature Biotechnology, Genome Biology, Genome Research, Nucleic Acids Research, Bioinformatics, Biology Direct, PLoS ONE

INSTITUTIONAL SERVICE

Co-Chair, The Annual Translational and Transformative Informatics Symposium (ATTIS), UAB, 05/2026